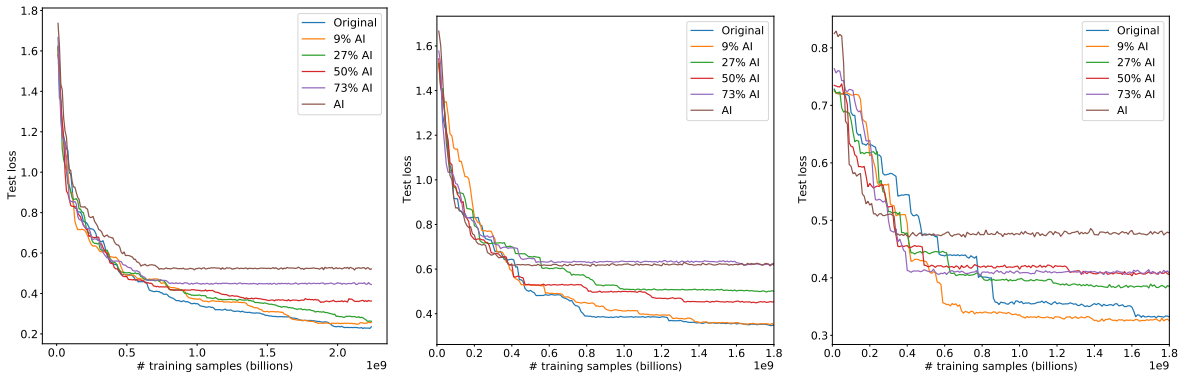

Figure 32. **Grokking in GCD Learning on mixed data.** Error losses of models trained on mixtures of clean and AI generated GCD data. 10 models. From left to right: base 4913, 2023 and 1000.

average curves over the 10 seeds one can discern a grokking-like delayed learning for the mixtures with relatively small amounts of AI data. This effect can be studied

The models used to generate the data were trained on about 300M examples, and correctly predict 22, 16 and 17 GCD below 100 for bases 1000, 2023 and 4913 respectively. We know (Table 3) that more training on AI-data data only will not improve those performances. On the other hand, we know that models trained on clean data will achieve larger performance. Specifically, out of 10 models trained on clean data, for base 1000, all 10 predict 23 GCD or more after 1.4B examples. The median number of examples needed for the models to predict 23 GCD or more is 465M. For base 2023, 7 models out of 10 predict 17 GCD or more after 2.1B examples. The median number of training samples after which the model bests a model trained on dirty data only is 530M. Finally, for base 4913, 9 clean models out of 10 predict more than 18 GCD after 1.7B examples. The median number of samples is 1.1B.

When zooming in to when the mixture models learn to predict GCD that are "unlearnable" with an AI-trained model, the grokking effect becomes more apparent.

Table 4 summarizes by listing the time (# of samples) when the mixture models finally learn a GCD that a purely AI-trained model cannot learn, and the delay (in millions samples) since the previous GCD was learned (see also Figure 30 to illustrate the comparison between the clean and the AI-trained model):

Table 4. **Samples until Mixture Models Learn a GCD that AI-trained Models Cannot Learn.** * small number of experiments

	Base 1000			Base 2023			Base 4913		
mixture rate	successes	samples (M)	delay	successes	sample (M)	delay	successes	samples (M)	delay
0% (clean)	10/10	465	243	7/10	530	567	10/10	1180	520
9%	8/10	560	320	8/10	715	530	9/10	910	340
27%	5/10	790	560	7/10	790	1220	10/10	1390	680
50%	2/10*	1310*	190*	7/10	1140	1220	8/10	1280	1180
73%	0	-	-	0	-	-	0	-	-

The delay period increases with increasing fraction of AI data in the mix. Thus, Table 4 clearly demonstrates the grokking effect of increasing plateau length with fraction of AI data, as predicted by our theory[7].

H. Details of Experiments with Llama2

In the realm of large language models (LLMs), the prevailing approach involves a pretraining and finetuning paradigm. For instance, GPT-3 undergoes pretraining on approximately 45TB of text data from diverse sources. This extensive pretraining endows it with a robust capability for a variety of downstream tasks, employing methods such as zero-shot learning, few-shot

[7]We were constrained to stop the experiments at after about 3B samples for most, due to heavy use of compute resources. This probably explains why for the larger AI-mixtures only a few experiments could successfully find new GCDs - the other experiments where still in the pre-grokking phase when they were stopped.

learning, or finetuning. Our study evaluates the phenomenon of model collapse in scenarios close to the contemporary 'synthetic data age.'

Utilizing one of the most advanced open-source models, `Llama-2 7B`, our research investigates the effects on LLMs when they undergo finetuning[8] with data generated by other LLMs. To ensure the generation of high-quality data and to provide a relevant but not trivial downstream task, we employ the Wikitext-103 dataset. We segment this dataset into chunks of 128 tokens, between each with a stride of 64 tokens, resulting in approximately 2.2 million chunks. Denote this dataset as $\mathcal{D}_0$. The task for generation involves producing the final 32 tokens given the initial 96 tokens from each chunk in the original dataset. In the initial generation (0-th generation), we use the `Llama-2 7B FT` model, which has been finetuned on $\mathcal{D}_0$, applying a generation loss that focuses solely on the cross-entropy loss of the final 32 tokens. We denote this initial model as $\mathcal{M}_0$, which demonstrates enhanced capacity for the generation task compared to the standard Llama-2 7B model. By querying $\mathcal{M}_0$ with the original 96 tokens from $\mathcal{D}_0$, we generate the dataset $\mathcal{D}_1$ and subsequently finetune `Llama-2 7B` on this dataset to obtain $\mathcal{M}_1$. This process is sequentially repeated to generate $\mathcal{D}_i$ from $\mathcal{M}_{i-1}$ and obtain $\mathcal{M}_i$ through finetuning. By comparing the performance of various $\mathcal{M}$ models on the test set derived from Wikitext-103, also segmented into 128-token chunks, we aim to investigate the model collapse in LLMs.

To prevent information leakage across chunks, we restrict the training to only include the loss on the final 32 tokens for all generations. Consequently, the models are never trained on the first 96 tokens coming from the original corpus. The size of the 2.2 million chunks can provide sufficient data for finetuning while avoiding overfitting, given the capacity of `Llama-2 7B`. Throughout the finetuning process, we maintain consistent settings using learning rate $5e^{-5}$ for LoRA, using Adam optimizer, dropout rate 0.1, trainable parameter fraction 0.062%. To eliminate the possibility of model collapse due to insufficient sampling and to gain insights into scenarios where more AI-generated data is produced than the model has been trained (or finetuned) on, we consistently utilize a model trained on half the dataset for generating subsequent datasets.

For completeness, we include Figure 33 with loss on the full chunks and Figure 34 that mix the generated data with original data. The mixing curve also aligns well with the grokking phenomenon predicted by theory.

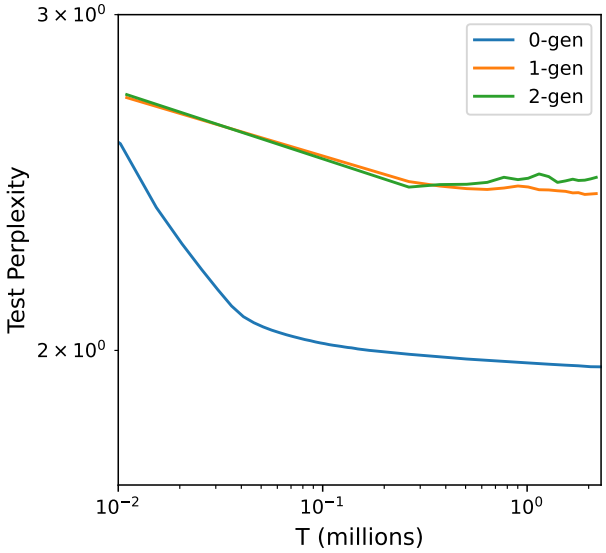

Figure 33. **Llama Generated Data.** Llama2 finetuning when the loss for training and evaluation is the cross-entropy for all tokens in the chunks, including the prompt.

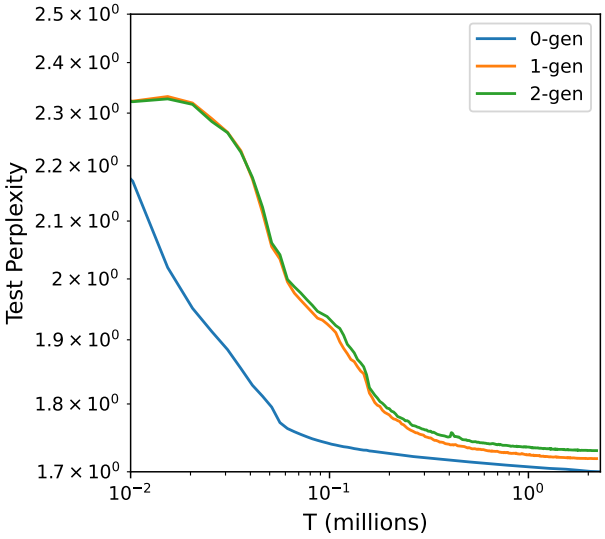

Figure 34. **Mixing Llama Generated Data with Original Data.** Similar setting as Figure 4 left. Starting from gen 1, we mix the generated data with the original one with a ratio of 90 to 10. Top-p^{inf} =0.9 and temperature τ =0.9.

[8]One can, in principle, replicate an experiment described here with training an LLM from scratch to demonstrate scaling law decay. We opted to not run such an experiment and instead focus on a more feasible finetuning setting, since just the language experiments described in the paper took weeks to run.

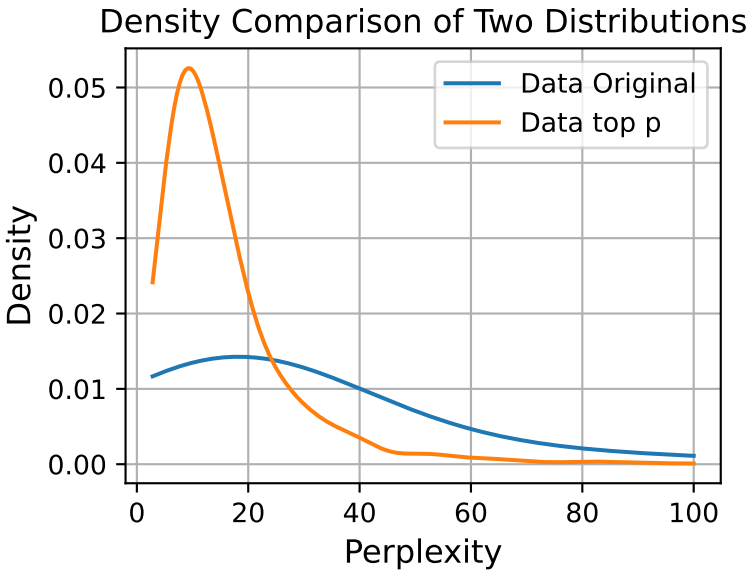

Figure 35. Sequential bigram data: top p^{inf} = 0.95 leads to similar effect as tail narrowing. 1000 data with sequence length 100.

I. More Studies on Tail Cutting and Tail Narrowing Effects

Here, we illustrate how tail cutting in the next-token distribution can lead to tail-narrowing for metrics that take the entire sequence into account, like perplexity. Figure 35 illustrates this for the autoregressive bigram model. This effect is likely due to the combinatorial factors we obtain when considering an additive (or multiplicative) measure like perplexity.